

Weekly Self-Study Tasks to Complete

Express-MongoDB Integration for Leaders

This document covers all of Week 5 of Phase 3 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote API lab, where you will connect the API to MongoDB with Mongoose. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep the lab project open so you can connect schemas and model calls to the Evergreen API.

Topic	Material Description	Est. Time	Link / Reference
MongoDB & CRUD	Foundations of Document Databases with MongoDB (Pinal Dave, Pluralsight, updated Nov 05, 2025; 2h 28m course). Watch “Introduction to MongoDB” (~22 min of the 36-min module). Documents, collections, and CRUD — the data layer an Express app talks to.	~22 min	pluralsight.com — Foundations of Document Databases with MongoDB
Mongoose Basics	The Mongoose “Getting Started” guide (official) — how to connect, define a schema, and use CRUD with a model.	~23 min	mongoosejs.com — Getting Started

Topic	Material Description	Est. Time	Link / Reference
Validation & Errors	The Mongoose “Validation” guide (official) – how required, enum, and min/max rules work and how validation errors surface.	~15 min	mongoosejs.com/docs/validation

Practice Activities (1 hour)

Complete this week’s hands-on lab: “Evergreen Insurance Quote API – Connecting Express to MongoDB.” You will define a Mongoose schema, replace in-memory data with database calls, and watch validation protect your data. The whole exercise takes about one hour.

In the lab you will:

- Open the Labfiles/evergreen-quote-express-mongodb folder and run npm install.
- Define the Quote schema in src/Quote.js with required, enum, and min/max rules.
- Seed the database with npm run seed, then start the server.
- Add the read routes using Quote.find() and Quote.findById().
- Add the create route using Quote.create() with the calculated premium.
- POST an invalid quote and watch schema validation return a 400 – the bad data is never saved.

The lab instructions (with “Why this matters” callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. The remaining Phase 3 weeks deploy and scale this service, so focus this week on schemas, models, and validation.

Topic	Material Description	Link / Reference
Mongoose	The Mongoose documentation — the authoritative guide to schemas, models, queries, validation, and middleware (Reference).	mongoosejs.com/docs
Node + Mongo	Databases for Node.js Developers (LinkedIn Learning) — a complete course building a database-backed API with Express (Video Course).	linkedin.com/learning — Node and MongoDB
Driver	MongoDB Node.js Driver documentation — the native driver underneath Mongoose, for when you need full control (Reference).	mongodb.com/docs/drivers/node